2

Relations

Chapter Learning Outcomes

CLO

2

- State the definition of binary relation and determine its domain and range.
- Represent a relation in a different ways.
- Determine the number of relations in a given set.
- Determine the type of a relation (reflexive, symmetric, transitive, equivalence).
- Perform the operations of union, intersection, difference, and composition on relations.
- Find powers of relations.

Relations

A **relation** is a pairing of input values with output values. It can be shown as a set of ordered pairs (x,y), where x is an input and y is an output.

The set of input (x) values for a relation is called the **domain**, and the set of output (y) values is called the **range**.

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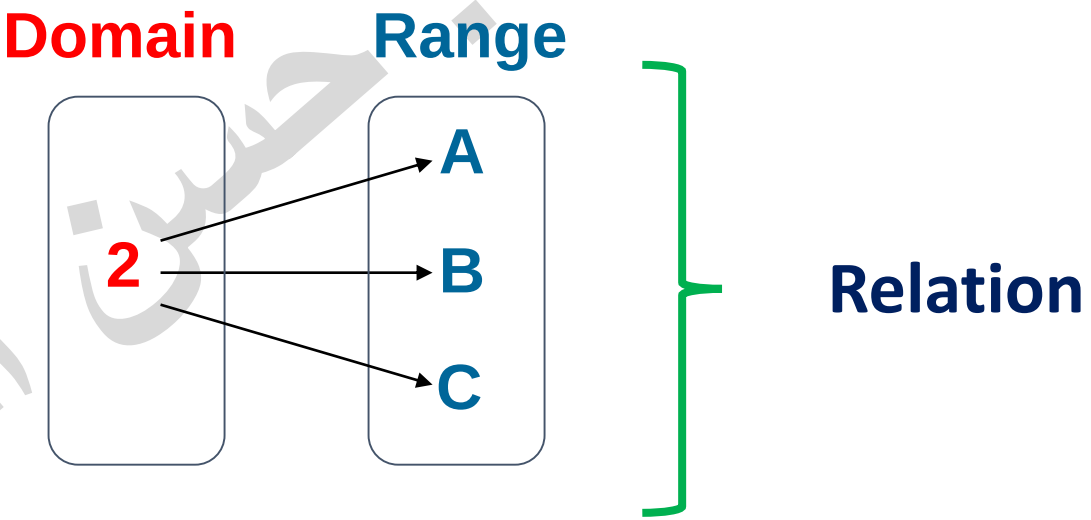
Mapping Diagram



A relation is a set of ordered pairs

R = {(2, A), (2, B), (2, C)}

$(x, y) \longrightarrow$ (domain, range)

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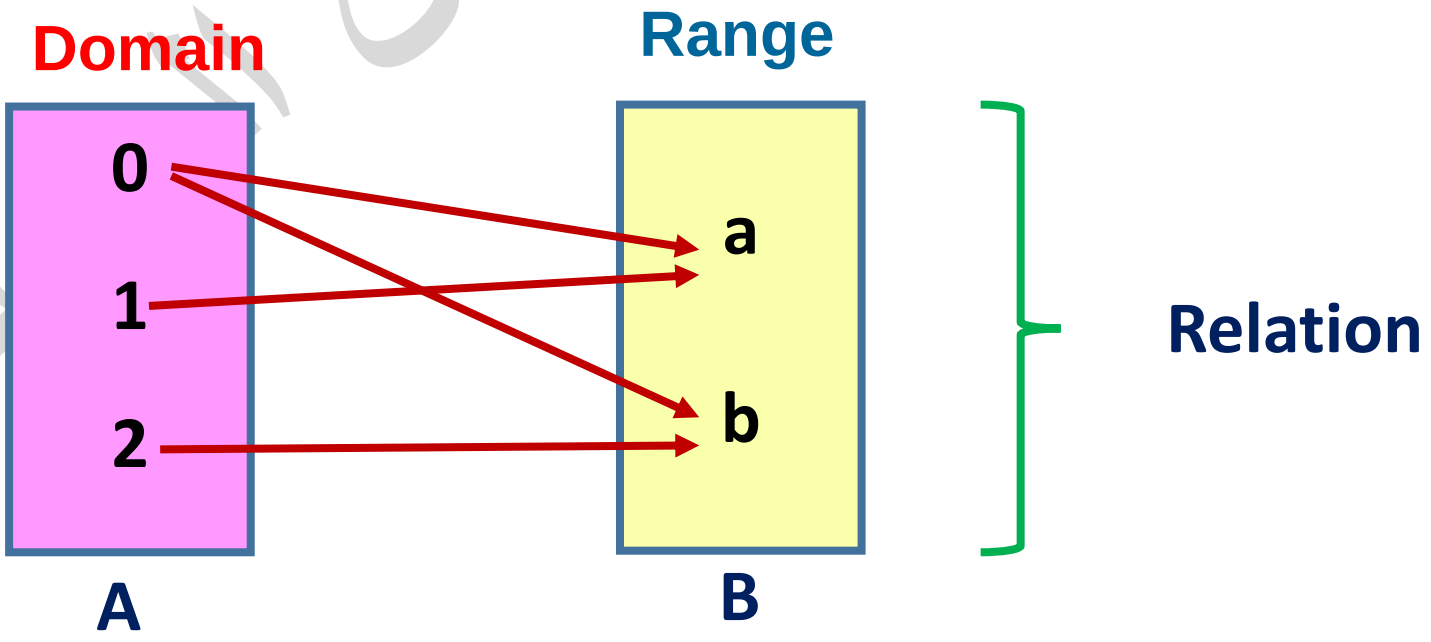
Example:

Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$ Then:

$\{(0, a), (0, b), (1,a), (2, b)\}$ is a relation from A to B.



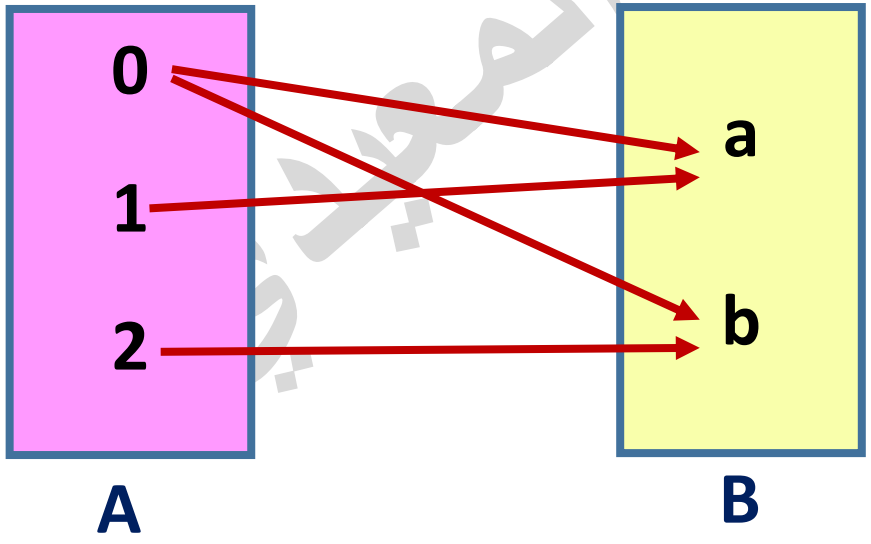
Represnt Relations

Example:

Let $A = \{0,1,2\}$ and $B = \{a,b\}$

R= $\{(0, a), (0, b), (1,a), (2, b)\}$ is a relation from A to B.

We can represent relations from a set *A* to a set *B* graphically or using a table:



R	a	b
0	1	1
1	1	0
2	0	1

Example

Give the domain and range for this relation:

{(100,5), (120,5), (140,6), (160,6), (180,12)}.

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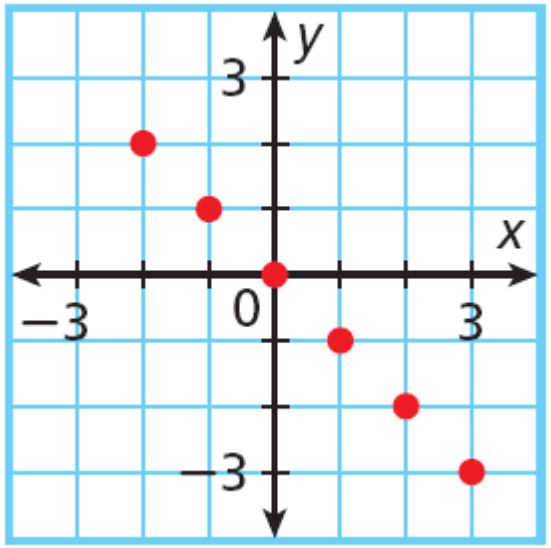
Example

Give the domain and range for the relation shown in the graph.



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Example

Give the domain and range for the following relation

$$R = \{(0,-5),(1,-4),(2,-3),(3,-2),(4,-1),(5,0)\}$$

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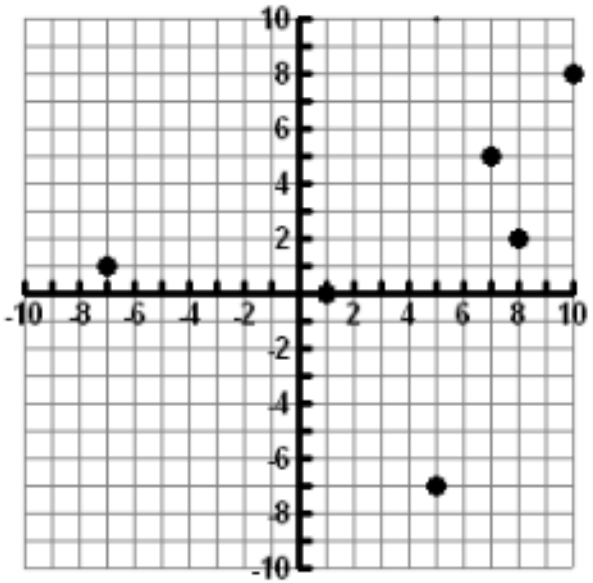
Relations

Example

Is the following function discrete or continuous? What is the Domain? What is the Range?



Type: _____

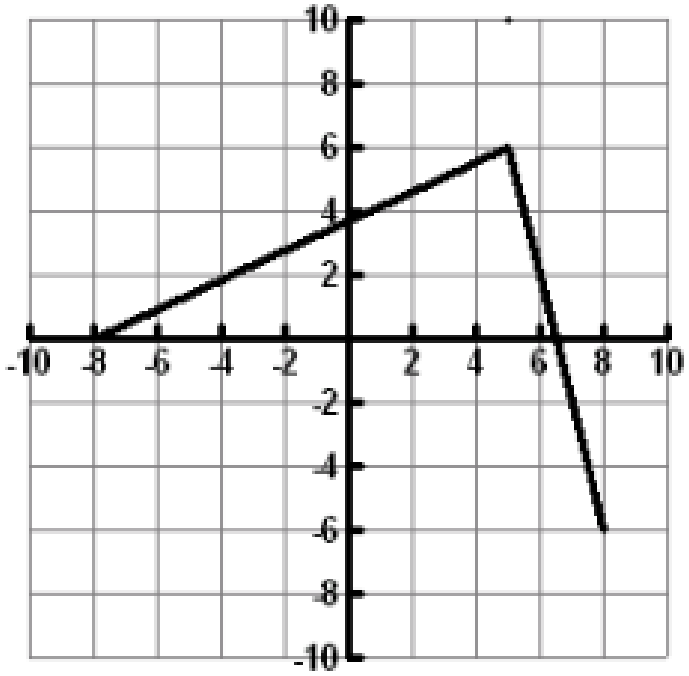
Domain: _____

Range: _____

Relations

Example

Is the following function discrete or continuous? What is the Domain? What is the Range?



Type: _____

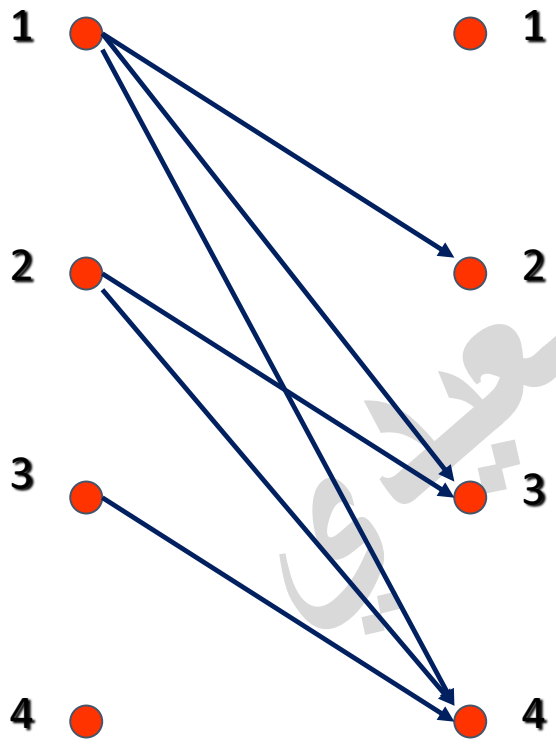
Domain: _____

Range: _____

Example

Let A = {1, 2, 3, 4}. Which ordered pairs are in the relation R = {(a, b) : a < b and a,b ∈ A } ?

Solution: R = {(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)}



R	1	2	3	4
1		1	1	1
2			1	1
3				1
4				

Relations

Example:

Suppose $A = \{1, 2, 3\}$ and $R = \{(a, b) : a, b \in A, a \leq b\}$.
Write all elements of R.

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Binary Relation on a Set

Definition: A binary relation R *on a set* A is a subset of $A \times A$ or a relation from A to A.

Example

- Suppose that $A = \{a,b,c\}$. Then $R = \{(a,a),(a,b),(a,c)\}$ is a relation on A.

Relations

Example:

Suppose A = {1, 2, 3, 4} and R = {(a, b) : a, b ∈ A, a divides b}.
Write all elements of R.

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No. of Relations on a Set

How many relations are there on a set A?

There are $2^{|A|^2}$ relations on a set A

Example Let B={1,2,3}, how many relations are there in B?

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No. of Relations on a Set

Example:

Assume a set A has 6 elements, how many relations can you generate from A?

$$\boxed{2^{|A|^2}}$$

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Relation Types

- Reflexive / Irreflexive
- Symmetric / Antisymmetric
- Transitive / Not Transitive
- Equivalence / Not Equivalence

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Reflexive / Irreflexive

Definition: A relation R on a set A is called **reflexive** if $(a, a) \in R$ for every element $a \in A$.

A relation *R* <u>on a set A</u> is called <u>reflexive</u> iff

$\forall (a,a) \in R$ for every element $a \in A$.

Definition: A relation on a set A is called **irreflexive** if

$\forall (a,a) \notin R$ for every element $a \in A$.

Reflexive / Irreflexive

EXAMPLE

Is the following relation on {1, 2, 3} reflexive/ irreflexive?

$R = \{(1,1), (1,2), (1,3), (2,1), (2,2), (3,3)\}$

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Reflexive / Irreflexive

EXAMPLE

Is the following relation on {1, 2, 3,4} reflexive/ irreflexive?

$R = \{(1,2), (1,3),(1,4),(2,4),(3,2),(4,1),(4,2)\}$

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Reflexive / Irreflexive

EXAMPLE

Are the following relations on {1, 2, 3} reflexive/ irreflexive?

relation	reflexive	irreflexive
R1 ={(1,1),(1,2),(2,1)}		
R2 = {(1,1), (1,2), (1,3), (2,1), (3,3)}		
R3 = {(1, 1), (2, 2), (3, 3)}		
R4 = {(1, 2), (2, 3)}		

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Quiz

Consider the following relations on {1,2,3,4}. Which are reflexive, irreflexive, neither?

- $R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$
- $R_2 = \{(2,4), (4,2)\}$
- $R_3 = \{(1,2), (2,3), (3,4)\}$
- $R_4 = \{(1,1), (2,2), (3,3), (4,4)\}$
- $R_5 = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$
- $R_6 = \{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$

Symmetric / Antisymmetric

Definition:

- A relation R on a set A is called **symmetric** if $(a, b) \in R \Rightarrow (b, a) \in R$.

Definition:

- A relation R on a set A is called **antisymmetric** if $(a, b) \in R$ and $(b, a) \in R$ then $a = b$.

Symmetric / Antisymmetric

EXAMPLE

Is the following relation on {1, 2, 3, 4} symmetric/antisymmetric?

$R = \{(1, 1), (1, 2), (2, 1), (3, 3), (4, 4)\}$

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Symmetric / Antisymmetric

EXAMPLE

Is the following relation on {1, 2, 3, 4} symmetric/antisymmetric?

R = {(1, 1), (2, 2), (3, 3),(4, 4)}

R	1	2	3	4
1	1			
2		1		
3			1	
4				1

Quiz

Consider the following relations on {1,2,3,4}. Which are symmetric, antisymmetric, both, or neither?

- $R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$
- $R_2 = \{(2,4), (4,2)\}$
- $R_3 = \{(1,2), (2,3), (3,4)\}$
- $R_4 = \{(1,1), (2,2), (3,3), (4,4)\}$
- $R_5 = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$
- $R_6 = \{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$

Transitive

Definition:

- A relation R on a set A is called **transitive** if whenever $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$.

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Transitive

EXAMPLE

Is the following relation on {1, 2, 3, 4} transitive?

$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$

Transitive

✓

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Quiz

Are the following relations on {1, 2, 3, 4} transitive?

relation	transitive
R1 = {(1, 1), (1, 2), (2, 2), (2, 1), (3, 3)}	
R2 = {(1, 3), (3, 2), (2, 1)}	
R3 = {(2, 4), (4, 3), (2, 3), (4, 1)}	

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Quiz

Consider the following relations on {1,2,3,4}. Which are transitive?

relation	transitive

- $R_1 = \{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$
- $R_2 = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$
- $R_3 = \{(2,4), (4,2)\}$
- $R_4 = \{(1,2), (2,3), (3,4)\}$
- $R_5 = \{(1,1), (2,2), (3,3), (4,4)\}$
- $R_6 = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$

Quiz

Consider the following relation on {1,2,3,4}. Name its properties?

$R = \{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\}$

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Quiz

Consider the following relation on {1,2,3,4}. Name its properties?

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

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Quiz

Consider the following relation on {1,2,3,4}. Name its properties?

$R = \{(1, 2), (2, 3), (3, 4)\}$

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Equivalence

Definition: A relation R on a set A is called **equivalence,** if it is **reflexive** and **symmetric** and **transitive.**

Consider the following relation on {1,2,3,4}. Check if it is equivalence or not?

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$$

Quiz

Consider the following relations on {1,2,3,4}. Name its properties?

$R_1 = \{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$

$R_2 = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4)\}$

$R_3 = \{(2,4), (4,2)\}$

$R_4 = \{(1,2), (2,3), (4,3)\}$

$R_5 = \{(1,1), (2,2), (3,3), (4,4)\}$

$R_6 = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$

Exercise

Determine the properties of each of the following relations defined on the set of all real numbers *R*:

- $R1 = \{(x,y) : x + y = 0\}$
- $R2 = \{(x,y) : x = y \vee x = -y\}$
- $R3 = \{(x,y) : x = 2y\}$

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$R1 = \{(x,y) : x + y = 0\}$

Solution:

- It is not reflexive since, for example, $(1,1) \notin R$.
- It is not irreflexive since, for example, $(0,0) \in R$.
- Since $x + y = y + x$, it follows that if $x + y = 0$ then $y + x = 0$, so the relation is symmetric.
- It is not antisymmetric since, for example, $(-1,1)$ and $(1, -1)$ are both in R, but $1 \neq -1$
- The relation is not transitive since, for example, $(1,-1) \in R$ and $(-1,1) \in R$, but $(1,1) \notin R$.

$R2 = \{(x,y) : x = y \vee x = -y\}$

Solution:

- Since for each x, xRx then it is reflexive.
- Since it is reflexive, it is not irreflexive.
- Since $x = \pm y$ if and only if $y = \pm x$, then it is symmetric.
- It is not antisymmetric since, for example, $(1,-1)$ and $(-1,1)$ are both in R but $1 \neq -1$
- It is also transitive because essentially the product of ± 1 and ± 1 is ± 1.

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$R3 = \{(x,y) : x = 2y\}$

Solution:

- It is not reflexive since, for example, $(1,1) \notin R$.
- It is not irreflexive since, for example, $(0,0) \in R$.
- It is not symmetric since, for example, $(2,1) \in R$ but $(1,2) \notin R$.
- It is antisymmetric because $x = y = 0$ is the only time that (x,y) and (y,x) are both in R.
- It is not transitive since, for example, $(4,2) \in R$ and $(2,1) \in R$ but $(4,1) \notin R$.

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Operations on Relations

Example: Let $A = \{1, 2, 3\}$, $B = \{u, v\}$

$R = \{(1, u),(2, u),(2, v),(3, u)\}$

$S = \{(1, v),(3, u),(3, v)\}$

Find $R \cap S$ **Find $|R \cap S|$**

$R \cap S =$

$|R \cap S| =$

Operations on Relations

Example: Let $A = \{1, 2, 3\}$, $B = \{u, v\}$

$R = \{(1, u),(2, u),(2, v),(3, u)\}$

$S = \{(1, v),(3, u),(3, v)\}$

Find $R \cup S$ **Find $|R \cup S|$**

$R \cup S =$

$|R \cup S| =$

Operations on Relations

Example: Let $A = \{1, 2, 3\}$, $B = \{u, v\}$

$R = \{(1, u),(2, u),(2, v),(3, u)\}$

$S = \{(1, v),(3, u),(3, v)\}$

Find R - S **Find |R - S|**

R - S =

|R - S| =

Operations on Relations

Example: Let $A = \{1, 2, 3\}$, $B = \{u, v\}$

$R = \{(1, u),(2, u),(2, v),(3, u)\}$

$S = \{(1, v),(3, u),(3, v)\}$

Find S - R **Find |S - R|**

S - R =

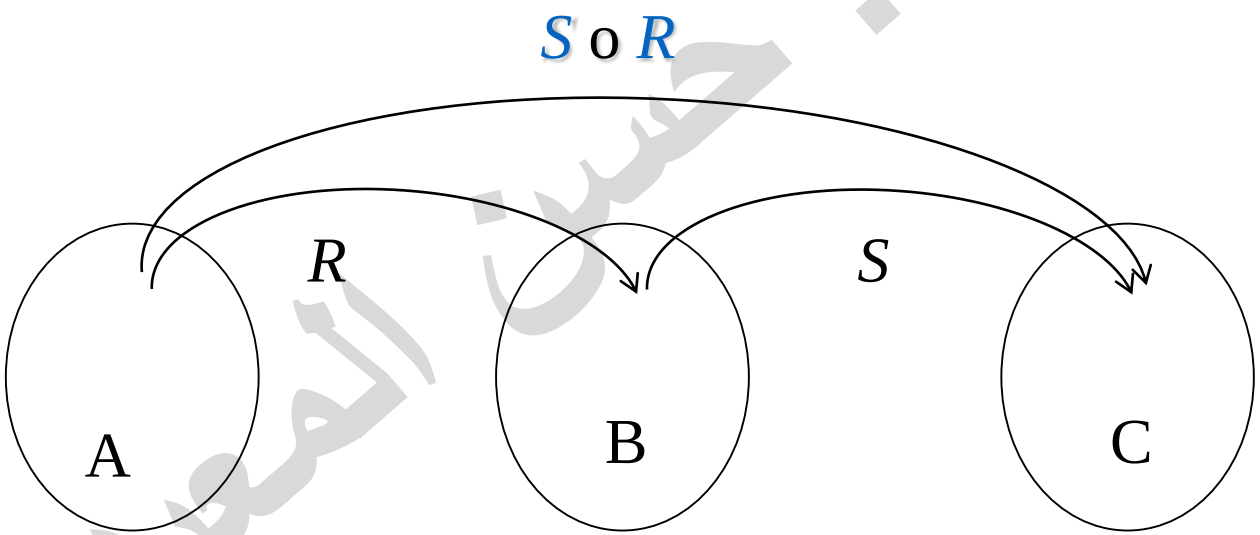
|S - R| =

Composition of Relations

- Let **R** be a relation from a set A to a set B.
- Let **S** a relation from B to a set C.
- The **composite of R and S** is the relation consisting of ordered pairs (a, c), where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$.
- We denote **the composite of R and S** by $S \circ R$.

S after R

Composition of Relations



S after R

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Composition of Relations

Example: Let $A = \{2, 4, 7, 9\}$ and R and S defined on A, where:

$$R = \{(2, 2),(4, 7),(7, 4),(9, 2)\}$$

$$S = \{(2, 9),(7, 4),(4, 2),(7,9)\}$$

Find R o S

S	R	R o S

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Composition of Relations

Example: Let $A = \{2, 4, 7, 9\}$ and R and S defined on A, where:

$$R = \{(2, 2),(4, 7),(7, 4),(9, 2)\}$$

$$S = \{(2, 9),(7, 4),(4, 2),(7,9)\}$$

Find S o R

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Composition of Relations

Example: Let $A = \{1, 2, 3\}$, $B = \{0, 1, 2\}$ and $C = \{a, b\}$

$R = \{(1, 0),(1, 2),(3, 1),(3, 2)\}$

$S = \{(0, b),(1, a),(2, b)\}$

Find S o R

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Composition of Relations

Example: Let $A = \{1, 2, 3\}$, $B = \{0, 1, 2\}$ and $C = \{a, b\}$

$R = \{(1, 0),(1, 2),(3, 1),(3, 2)\}$

$S = \{(0, b),(1, a),(2, b)\}$

Find R o S

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Composition of Relations

Example

- Let $A=\{1, 2, 3\}$, $B=\{x, y, w, z\}$, $C=\{0, 1, 2\}$,
- R a relation from A to B:
- $R = \{(1, x), (1, z), (2, w), (3, x), (3, z)\}$, and
- S a relation from B to C:
- $S = \{(x,0), (y,0), (w,1), (w,2), (z,1)\}$

What is $S \circ R$?

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Quiz

Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d\}$ and $C = \{x, y, z\}$

$R = \{(1, a),(2, d),(3, a),(3, b),(3, d)\}$

$S = \{(b, x),(b, z),(c, y),(d, z)\}$

Find S o R

Find R o S

Power of Relations

Definition:

Let R be a relation on the set A. The powers R^n, $n = 1, 2, 3,\ldots$ are defined recursively by:

- $R^1 = R$,
- $R^{n+1}=R^n \circ R$, for $n \geq 1$.

So:

- $R^2 = R \circ R$,
- $R^3 = R^2 \circ R$
- $R^4 = R^3 \circ R$
- …

Powers of Relations

Example: Let $A = \{1, 2, 3, 4\}$, $R: A \rightarrow A$ *such that:*

$$R = \{(1, 2),(2, 3),(2, 4),(3, 3)\}$$

Find R^1

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Powers of Relations

Example: Let $A = \{1, 2, 3, 4\}$, $R: A \rightarrow A$ *such that:*

$$R = \{(1, 2),(2, 3),(2, 4),(3, 3)\}$$

Find R^2

R	R^1	R o R

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Powers of Relations

Example: Let $A = \{1, 2, 3, 4\}$, $R: A \rightarrow A$ *such that:*

$$R = \{(1, 2),(2, 3),(2, 4),(3, 3)\}$$

$$R^2 = \{(1, 3),(1, 4),(2, 3),(3, 3)\}$$

Find R^3

R	R^2	R^2 o R

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Powers of Relations

Example: Let $A = \{1, 2, 3, 4\}$, $R: A \rightarrow A$ *such that:*

$$R = \{(1, 2),(2, 3),(2, 4),(3, 3)\}$$

$$R^3 = \{(1, 3),(2, 3),(3, 3)\}$$

Find R^4

R	R^3	R^3 o R

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Quiz

Consider the following relations on {1,2,3,4}.

- $R = \{(2,2), (2,3), (2,4), (3,2), (3,3), (3,4)\}$

- Find R^2
- Find R^3
- Find R^4

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Check - Chapter Learning Outcomes

CLO

2

- State the definition of binary relation and determine its domain and range.
- Represent a relation in a different ways.
- Determine the number of relations in a given set.
- Determine the type of a relation (reflexive, symmetric, transitive, equivalence).
- Perform the operations of union, intersection, difference, and composition on relations.
- Find powers of relations.

End of Chapter

2